

Moveit Cartesian Path

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1. Environment

Ubuntu 22.04

ROS2: Humble

2. Driving the Real Machine

Driving the real machine involves subscribing to Moveit2's `/joint_states` topic to convert the joint state information of the robotic arm into control data for the actual robotic arm.

Note: Since the real robotic arm lacks obstacle avoidance, some poses may encounter obstacles; therefore, the planned robotic arm movements should be as reasonable as possible and avoid locations with obstacles.

(It is recommended to use preset poses for demonstrating the real machine.)

2.1 Starting the Real Machine

If not driving the real machine, demonstrate the robotic arm movements in MoveIt simulation (skip the following steps):

```
ros2 run dofbot_driver dofbot_driver
```

2.2 Starting MoveIt2

```
ros2 launch dofbot_moveit demo.launch.py
```

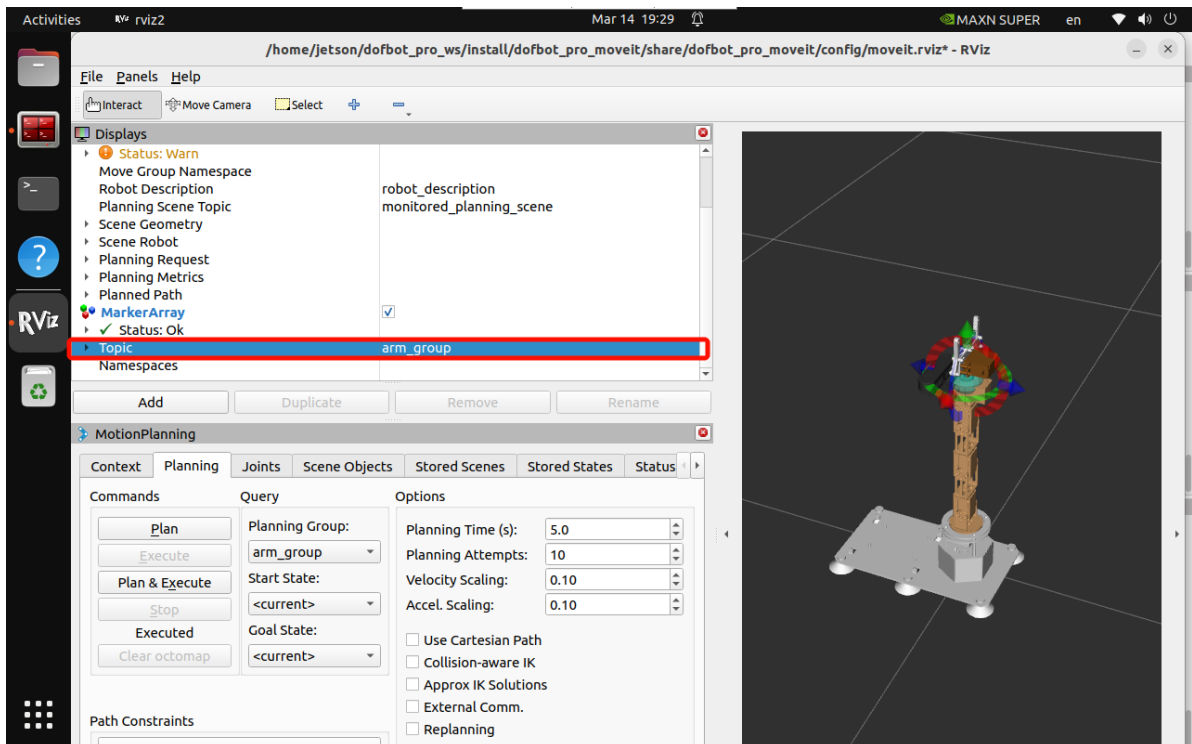
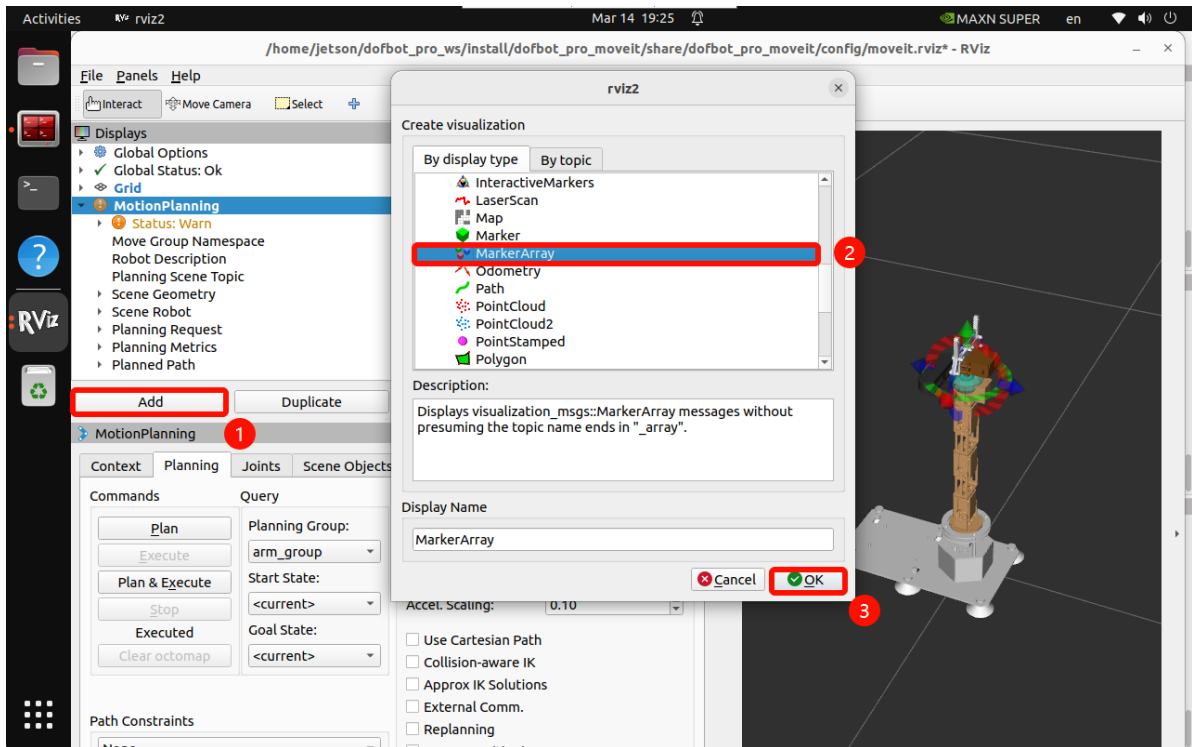
3. Cartesian Path

The Cartesian path refers to the linear motion path of the robotic arm's end effector in the Cartesian coordinate system.

Due to the limitations imposed by the robotic arm's degrees of freedom and structural design, identifying suitable Cartesian waypoints can be extremely challenging.

3.1: Visualization

Before executing the launch command, you must add the `MarkerArray` plugin within `RViz2` to visualize the straight-line path generated by the Cartesian planner. For the `MarkerArray` plugin, be sure to select the `arm_group` topic.



3.2: Launch Command

Ensure that the robotic arm has been successfully loaded into MoveIt and that the message `You can start planning now!` has appeared. Then, execute the following command; the robotic arm will proceed to plan its Cartesian path autonomously:

```
ros2 run dofbot_moveit cartesian_path
```

